

Project Proposals on Algorithms with Predictions

- 1) [Writing] Write lecture notes for algorithms with predictions. Besides what covered in class, these should contain a coverage of paging for possible classroom use.
- 2) [Writing] So far about a dozen papers, as well as a patent¹, investigated how to improve paging using ML predictions. Write a survey on paging with predictions highlighting the key ideas and techniques, experimental results, and the research questions that remain open.
- 3) [Study/Implementation/Open Problem] Dinitz et al. [6] initiate the study of algorithms with *distributional* predictions, where the prediction is a distribution instead of a single value. They initiate this line of research with binary search as a case study. Study and present their paper. Can you improve over their theoretical or experimental results?
- 4) [Study/Implementation/Open Problem] Last year I proposed to investigate the ski rental problem in the setting of distributional predictions. No one picked that project. Somebody else worked on it, and just a few weeks ago uploaded a preprint showing (essentially) tight bounds: <https://arxiv.org/abs/2602.21104v1>. Study and present their paper. Can you simplify their analyses? Can you run an experimental evaluation of their algorithm? What about other problems in the setting of distributional predictions?
- 5) [Study/Implementation/Open Problem] The algorithms-with-predictions framework has been used extensively to design online algorithms with improved competitive ratios. Another possibility is to use predictions to improve the running time of *offline* algorithms: in fact, while in classic algorithmics we make the assumption that problems are solved from scratch, in real applications many problems are solved repeatedly on similar input instances, thus opening the opportunity to design better algorithms by leveraging similarities across problem instances. Researchers have recently developed such accelerated algorithms for several problems, such as shortest paths [3, 7], maximum flow [5], and minimum cut [8], or to obtain better approximation algorithms [1, 9]. Study and present one of those papers. Possibilities for research include improving over these results or investigating other problems.
- 6) [Study/Implementation/Open Problem] Data structures can also benefit from applying ML predictions, giving better performance than their classical counterparts that do not take advantage of patterns in the input data. Study and present [2], which shows how to improve heaps. Possibilities for research include improving over these results or investigating other data structures.

¹<https://www.freepatentsonline.com/20250077441.pdf>

- 7) [Study/Implementation/Open Problem] Pick your favorite problem and 1) study the relevant paper(s) from the list in Further Material², 2) do an experimental evaluation of algorithms with predictions for that problem (like that in [4] for paging), or 3) solve an open problem suggested by any of the papers from the list in Further Material. If nobody ever studied your favorite problem under the lens of algorithms with predictions, be the first to do so!

References

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²<https://algorithms-with-predictions.github.io/>